



Shyam Banerjee
Bachelors in Statistics
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Course	College/University	Year	CGPA/%
Bachelors in Statistics	Indian Statistical Institute	2024-27	82.0
Intermediate/+2	Gitanjali Public School	2024	94.8
High School	Gitanjali Public School	2022	97.4

SCHOLASTIC ACHIEVEMENTS

- Secured **rank 9** in Madhava Mathematics Competition [2026]
- Secured **rank 48** in Simon Marais Mathematics Competition [2026]
- Finalist at Trilytics Case Analytics (IIM Calcutta) [2025]
- Cleared IOQM (Regional) [2022]

WORK EXPERIENCE

- Software Engineer Intern** | Alumnus Software Ltd. [Apr'26-Sep'26]
- Summer School in Astrostatistics** | Penn State University [Jun'25]

SKILLS & INTERESTS

- Programming Languages:** Python, C++, R
- Tools & Libraries:** NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, Gaussian Processes, $\text{\LaTeX}$
- Interests:** Statistical Inference, Astrostatistics, Time Series Analysis, Machine Learning, Quantitative Finance, Probabilistic Modelling

PROJECTS

- GP-based Imputation in Kepler Stellar Light Curves** | *Independent Project* [May'26-Jun'26]
 - Implemented Gaussian Process regression with RBF kernels for probabilistic modeling and imputation of noisy stellar light curves using Python, NumPy, SciPy, Matplotlib, and Scikit-learn.
 - Simulated noisy periodic light curves and evaluated GP-based recovery under missing-data regimes using Kepler observations of both quiet and rotationally variable stars.
 - Performed posterior inference for periodic stellar signals using harmonic models, comparing inferred structure across stationary and nonstationary variability patterns.
 - Inspired by lectures on Gaussian Processes by Dr. Suzanne Aigrain
- Non-stationary Time Series Modeling** | **Dr. Shyamal Krishna De** [Apr'26-May'26]
 - Developed a seasonal ARIMA forecasting pipeline for weekly PM2.5 concentration data from CPCB Delhi using ~ 8 years of observations.
 - Performed nonstationarity diagnosis using ADF, KPSS, Mann-Kendall, and Seasonal Mann-Kendall tests, followed by Box-Cox variance stabilisation and iterative seasonal/regular differencing.
 - Implemented forecasting and residual diagnostics in R using `forecast`, `tseries`, and `urca`; evaluated probabilistic forecast coverage on held-out test data.
 - Coursework project for Statistical Methods 4

RELEVANT COURSES

- Statistical Inference** | **Linear Statistical Models** | **Probability Theory** | **Stochastic Process** | **Statistical Methods** | **Linear Algebra** | **Design and Analysis of Algorithms** | **Differential Equations** | **Sample Survey** [B.Stat]

VOLUNTEERING EXPERIENCE

- Mentoring** | **AP Tutors** [Sep'25-Mar'26]
 - Guided students for AP Stats and SAT Maths

EXTRACURRICULAR ACTIVITIES

- Competitive Programming (Codeforces)
- Game Theoretic Puzzles
- Sudoku, Mahjong